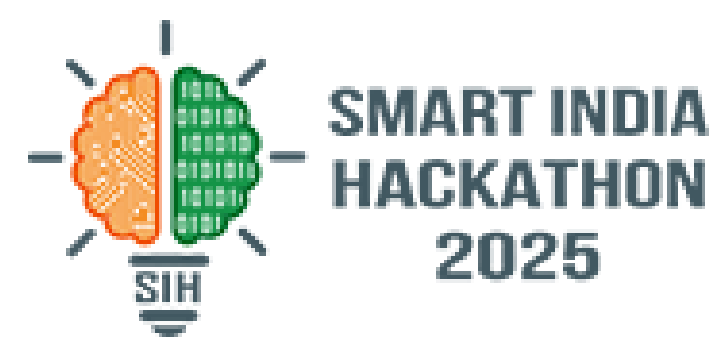


SMART INDIA HACKATHON 2025



PROBLEM STATEMENT ID – 25037

**PROBLEM STATEMENT TITLE – Development
of a low-cost camera-based automated beach
sand grain size mapping system**

THEME – Miscellaneous

PS CATEGORY – Hardware

TEAM ID – 86209

TEAM NAME – Dune It Better



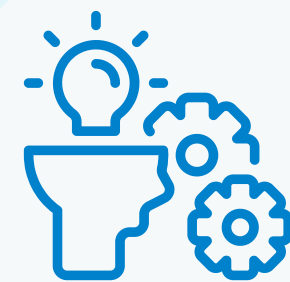
EXPECTED OUTCOME

- A **low-cost camera-based automated mapping** system for estimating the **sediment grain size distribution** of sandy beach areas (berm, intertidal, dune).
- Integration of a **GNSS/GPS receiver** to accurately map the **geographic position** of each measurement location.
- An **automated image processing algorithm** for **beach sand grain size estimation and classification**.
- **Validation using** sample collected from the area of interest.

HOW IT ADDRESSES THE PROBLEM

- **Eliminates manual sampling & lab analysis**, replacing them with quick, **on-site digital measurements**.
- Enables **rapid and repeatable monitoring**,
- Captures real-time changes in dynamic **beach sediments** after **tides, storms, or seasonal shifts**.
- Supports large-scale coastal monitoring by offering a **low-cost, portable** solution deployable across multiple sites.

OUR IDEA



A **low-cost portable camera system** integrated with GNSS/GPS.

Captures sand images at **~42 cm height (75° FOV)**.



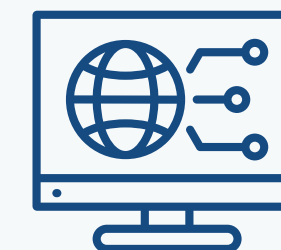
Mounted on a **tetrapod**, a four legged stand with 1ft x 1ft base markers for standardized imaging.

Geo-tagged results stored and visualized with GPS integration.



Applies **marker-based reorientation, pixel-to-length calibration, and uniform re-illumination algorithm**.

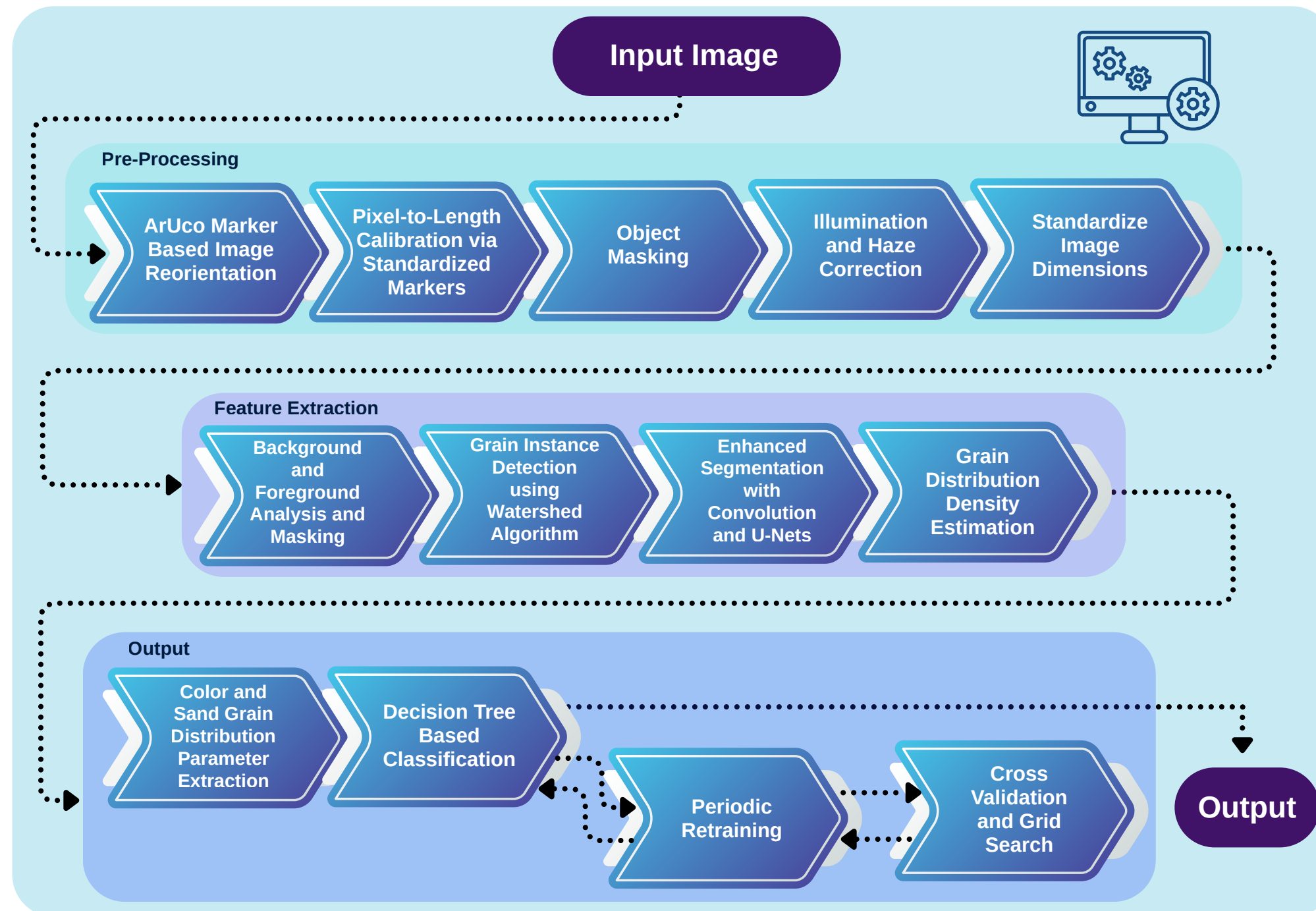
Processed via **cloud-based ML pipeline** → pixel-to-mm conversion, segmentation, **grain size distribution**, and classification into **beach categories (berm, intertidal, dune)**.



UNIQUENESS

- **Standardized Imaging:** A fixed **tetrapod base (1ft x 1ft markers)** ensures repeatable image capture from a constant 42 cm height, **removing the need for reference objects**.
- **ArUco markers** for precise image reorientation and accurate scaling.
- **LED strips on tetrapod legs** to prevent stand shadows on sand under sunlight.
- **Re-illumination & haze correction algorithm** for consistent image quality in varying light conditions.
- **Dynamic Dual-Mode Storage:** The system automatically switches between **local SD card storage** when offline and **cloud-based upload/processing** when Wi-Fi is available.
- **Automated Workflow:** Fully automated process replaces manual sampling and lab analysis with quick on-site digital measurements.

AI/ML processing



GPS/GNSS



Sentinel-2

- To deal with changing coastlines
- 10m × 10m resolution** images are captured for each coastline **around the world**.
- Updated every **10 days** to help identify areas that need to be mapped again.

Handheld device components

 GNSS sensor for geo-tagging	 Wi-Fi module for cloud
 R-Pi Camera module 3	 Raspberry Pi 5 Micro Controller
 22.5 Watt Power source	 SD Card for buffering /storage

Technologies/
applications used

- Fusion 360**, CAD design software –Used for making the hand held prototype .
- Raspberry Pi** – used as low cost, versatile and compact processor.
- Pre-processing – happens on Raspberry Pi itself.
- Python IDE, Code editor.
- Google collab**, Cloud notebooks
- Deep learning models (cloud), were trained on these two tools.
- ML/AI pipe-lining**, enables us an automated workflow.
- Camera module**, for image capture.
- GNSS sensor** (geo-tagging), enables to get current live position.
- WiFi module** (communication medium), both sensors are interfaced with Raspberry Pi.
- Cloud-connected to pre-processing part through WiFi module.

Limitation of Current Solution

The **AI model's accuracy** is highly dependent on its training data. The model may perform poorly on **images with environmental variability**, such as wet sand, shadows, different lighting conditions, or the presence of debris like shells.

The system uses a **standard GPS module** to log the coordinates of each measurement and **stores the data locally** on an SD card when offline.

This lack of an **intuitive Graphical User Interface (GUI)** makes the system difficult for non-technical users to operate and get real-time feedback. It **limits the system's potential** for widespread adoption in citizen science projects.

Proposed Solution of Limitation

The model should be trained on a diverse dataset of beach images captured in varied real-world conditions. Using a **citizen science approach**, users can contribute data by following a **defined protocol**, ensuring consistency while continuously improving the model's **robustness and accuracy**.^[1]

This approach leverages **data from many users** to reduce positional errors. By **averaging multiple data points** from similar locations, it improves accuracy while remaining **scalable and cost-effective**.^[2]

Develop a simple, user-friendly GUI using a **small touchscreen** connected to the **Raspberry Pi**. The interface would display a live camera feed, show the **classification results in real-time**, and provide an on-screen menu.

Bill of Materials

Raspberry Pi 5:	6240
Camera Module:	2800
GPS module:	230
Power Bank:	799
SD card(32GB):	479
Push Button:	50
LED strip:	224
Total Cost:	₹ 10864

For Components link: [Click here](#)

Feasibility

Raspberry Pi 5 can run **TensorFlow Lite** models at speeds similar to a dedicated Coral TPU, demonstrating its capability for **on-device machine learning tasks**.^[3]

Technical

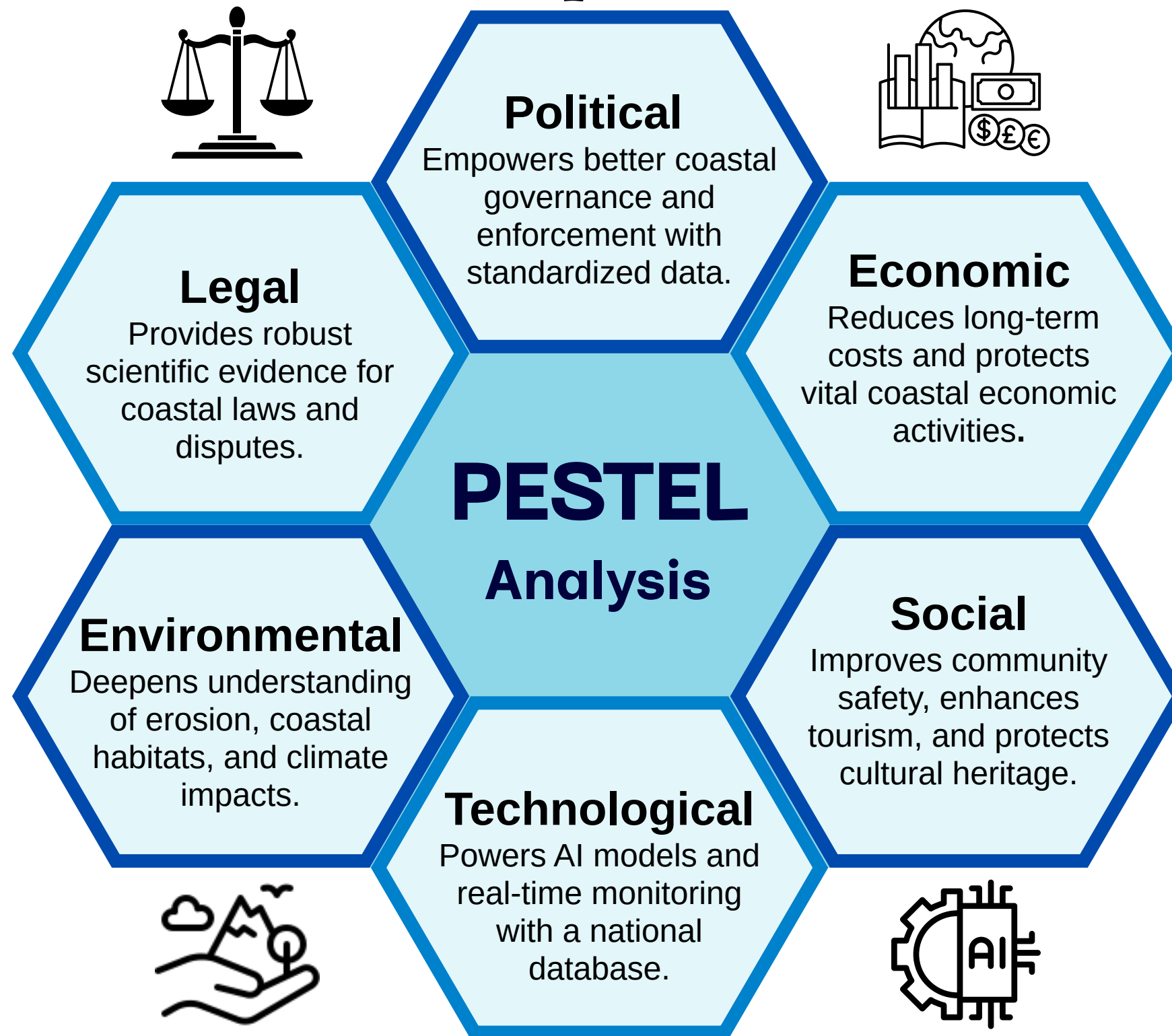
Automated systems, while having a higher initial setup cost, can **save significant time and labor** compared to manual analysis, and they offer **greater consistency and scalability** for long-term use.

Economic

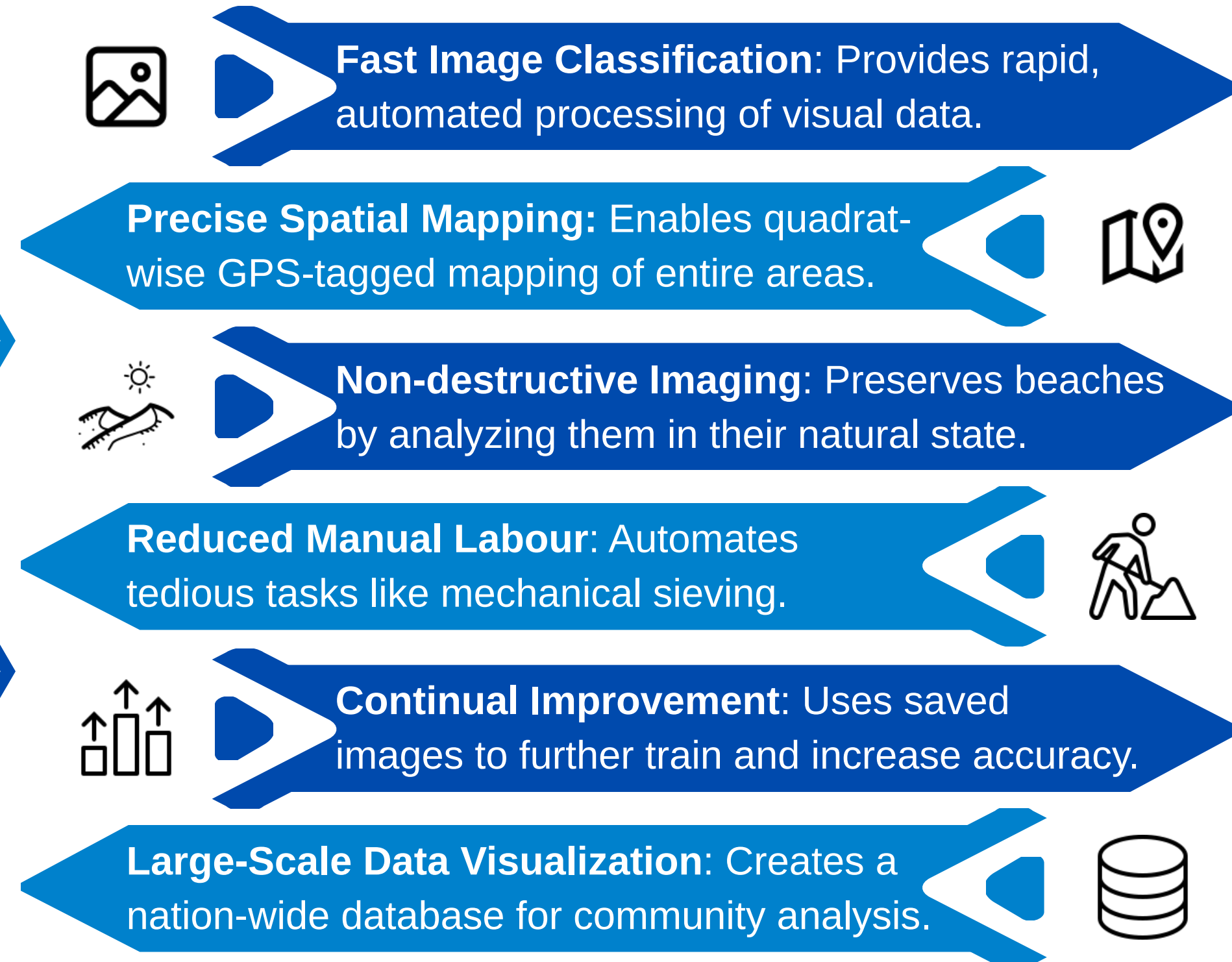
Continuous improvement loop: The AI model will be periodically retrained with new, ground-truth data obtained from **traditional sieve analysis**, ensuring its long-term accuracy and adaptability to new environments.

Operational

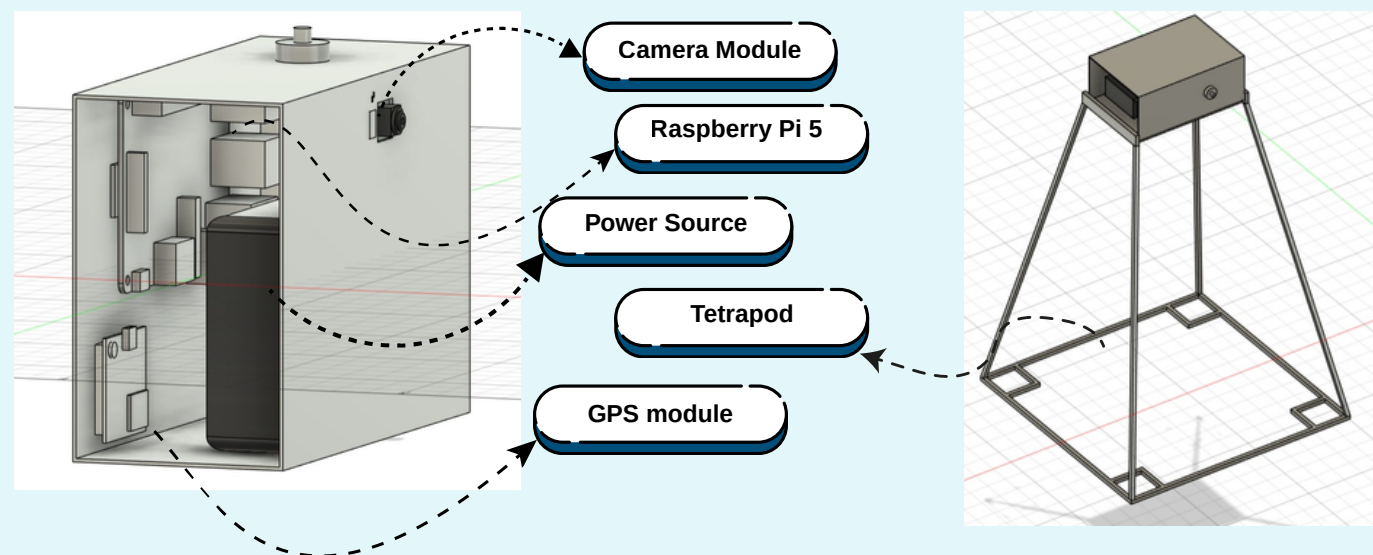
Impacts



Benefits



3D Model of the Prototype

[Link](#)

REFERENCE PAPERS

1. <https://www.mdpi.com/2077-1312/13/6/1108>
2. <https://www.mdpi.com/2072-4292/13/14/2829>
3. https://www.politesi.polimi.it/retrieve/a81cb05c-6352-616b-053-1605fe0a889a/2017_07_Marchetti.pdf
4. https://www.nuce.psu.edu/motta/Publications/188_Xu_JNM_2024.pdf
5. [An Effective Approach to Improving Low-Cost GPS Positioning Accuracy in Real-Time Navigation](#)

REFERENCE ARTICLES

1. <https://sandsnap-erdchl.hub.arcgis.com/>
2. https://docs.opencv.org/4.x/d3/db4/tutorial_py_watershed.html
3. <https://www.geoengineer.org/education/laboratory-testing/step-by-step-guide-for-grain-size-analysis>
4. [https://www.researchgate.net/publication/375667789_Automated Grain Boundary Detection for Bright-Field Transmission Electron Microscopy Images via U-Net](https://www.researchgate.net/publication/375667789_Automated_Grain_Boundary_Detection_for_Bright-Field_Transmission_Electron_Microscopy_Images_via_U-Net)
5. <https://www.hackster.io/news/benchmarking-tensorflow-and-tensorflow-lite-on-raspberry-pi-5-b9156d58a6a2>

Sample Decision Tree Classifier Performance Metrics

Decision Tree was chosen after comparing the performance metrics among **Decision Trees, Support Vector Machines and Logistic Regression**. The Dataset contained about **3000 samples** and more are being acquired through the web and **class imbalance** is mitigated through **Smote oversampling** to improve model's **generalization and performance**.

type \ metric	Precision	Recall	F1
Berm	0.87	0.8	0.83
Intertidal	0.72	0.76	0.74
Dune	0.5	0.67	0.57
Macro Avg	0.7	0.74	0.72
Weight Avg	0.79	0.78	0.79

Table: Test Split Metrics

Our Prototype ML Pipeline & Classifier Code: [GitHub Link](#)